RTTI - One way streets

Introduction to semantic modelling

A **concept scheme** (in SKOS: *ConceptScheme*) is a structured collection of concepts that carry a fixed, shared meaning within a particular domain. The goal is for everyone working with the model — people and systems alike — to use the same words for the same things. That sounds obvious, but in practice language causes a surprising amount of confusion: what one organisation calls a *measurement point*, another calls a *sensor*, a *location*, or an *observation station*.

A concept scheme solves this by recording, for each concept:

- a **preferred label** (prefLabel) it is known by,
- a **definition** that unambiguously describes what the concept does and does not cover,
- and optionally a **scope note** or **example** to clarify borderline cases.

Concepts rarely stand on their own. They are related **hierarchically** (a vehicle is broader than a truck) or **associatively** (a road user relates to a road segment). These relationships make it possible for queries and analyses to navigate between general and specific concepts, and to make connections between concepts explicit. Within SKOS we use `broader`, `narrower`, and `related` for that.

When a concept also exists elsewhere — in a national or European standard, for instance — we record that as a **mapping relation**. This prevents reinventing the wheel: we link to the source so interoperability with external systems remains possible.

This document describes the concept scheme as it is currently recorded. The following sections list the metadata, every concept, the relationships between them, and — where the model is connected to a data source — how it is linked to that source data.

RTTI - One way streets

Description

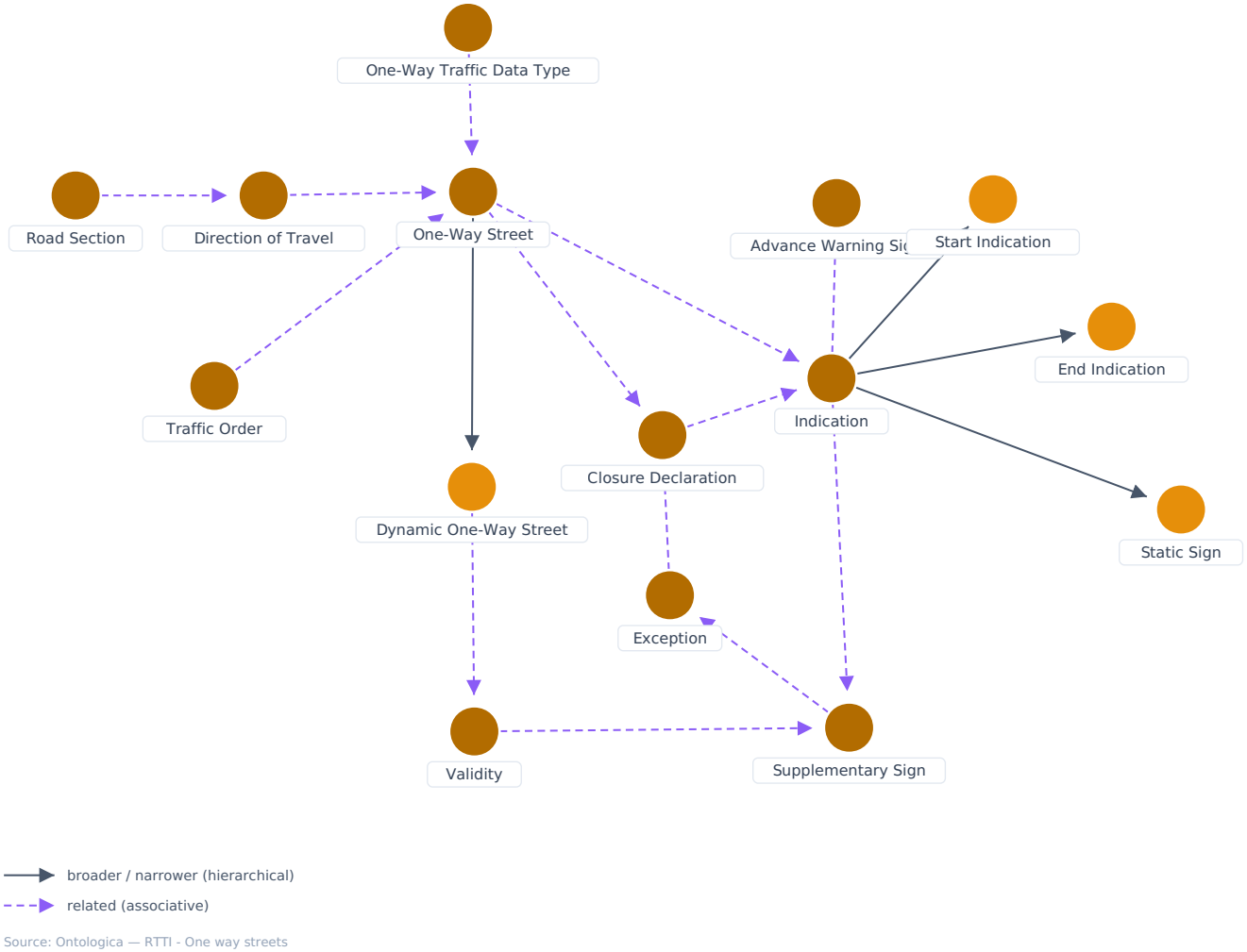
This vocabulary describes the information needed to record where traffic may travel in only one direction. The core is the permitted direction of travel for a street: on a one-way street you may only go one way. Exceptions are common — a very frequent one is 'except cyclists', allowing cyclists to travel both ways, or exceptions for line buses or local destination traffic. Sometimes the one-way rule applies only at certain times (for example during rush hour or events); this is called a dynamic one-way street. For each one-way situation, the vocabulary also records: where it applies (which stretch of road) and in which direction, how road users see it (through roadside traffic signs, supplementary signs showing exceptions or times, and advance warning signs), when it applies (always or only at certain times), who any exceptions apply to, and the official decision the rule is based on. By describing all of this in a consistent way, different parties — such as road authorities, navigation systems and travel information services — can reliably exchange and use the information. This matters especially for route planning: a cycling route needs to know a street is often usable both ways, while a car or freight route must respect the one-way rule. The vocabulary is part of the European arrangements for travel and traffic information (the ITS Directive 2010/40/EU and the RTTI Regulation (EU) 2022/670) and connects to existing Dutch sources for roads and traffic signs (NDW and the National Road Database).

Metadata

Field	Value
URI	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064
Version	0.5.0
Publication status	Draft (Under development)
Management status	Active
Organization	Nationaal Toegangspunt Mobiliteitsdata
Created on	2026-02-27
Last modified on	2026-10-06
Last modified by	arthur.keesen@ndw.nu

Model overview

Visual overview of the concepts and their relations. Hierarchical relations are drawn with solid arrows; associative relations are drawn with dashed arrows.



Source: Ontologica — RTTI - One way streets

Concepts

One-Way Traffic Data Type

Concept details

Field	Value
Definition	Data type within the RTTI context describing the set of concepts required to unambiguously record and exchange one-way traffic regulations on the Dutch road network. Encompasses the direction of travel, the indication, any exceptions and the legal basis.

One-Way Street

Concept details

Field	Value
Definition	A road segment where traffic is allowed to move in only one direction.

Road Section

Concept details

Field	Value
Definition	A section of road between two consecutive junctions or intersections, or – if no side road is present – between two points to which a traffic measure applies. In the National Road Database (NWB) a road section is the smallest unit for which road attributes are registered.

Direction of Travel

Concept details

Field	Value
Definition	The direction in which traffic moves on a carriageway or lane. For one-way streets, this attribute unambiguously encodes the permitted direction of travel.

Closure Declaration

Concept details

Field	Value
Definition	A closure declaration prohibiting traffic from entering a road segment in a specific direction. Indicated by sign C2 (RVV 1990): one-way street, closed in this direction for vehicles, riders and handlers of draught or pack animals or livestock.

Indication

Concept details

Field	Value
Definition	The traffic sign or combination of signs by which a one-way traffic regulation is communicated to road users.

Static Sign

Concept details

Field	Value
Definition	A physical ('tin') traffic sign permanently placed to indicate the one-way traffic regulation.

Start Indication

Concept details

Field	Value
Definition	The traffic sign indicating the point at which the one-way regulation takes effect.

End Indication

Concept details

Field	Value
Definition	The traffic sign or point at which the one-way regulation ceases to apply.

Supplementary Sign

Concept details

Field	Value
Definition	A sign mounted below a traffic sign to provide additional specification of the scope of that traffic sign.

Advance Warning Sign

Concept details

Field	Value
Definition	An advance warning sign placed upstream of the one-way street, at a junction where road users have the opportunity to choose an alternative route.

Exception

Concept details

Field	Value
Definition	A condition that modifies the standard one-way regulation, allowing certain road user categories to travel in the otherwise closed direction. Exceptions are communicated via supplementary signs.

Validity

Concept details

Field	Value
Definition	The time period during which the one-way regulation applies.

Dynamic One-Way Street

Concept details

Field	Value
Definition	A road segment where the one-way regulation is temporary or changeable, imposed in response to time of day, events, or prevailing conditions.

Traffic Order

Concept details

Field	Value
Definition	A formal traffic order constituting the legal basis for imposing a one-way traffic regulation.

Relations per concept

Outgoing semantic relations only — these stay inside the model.

One-Way Traffic Data Type

Semantic relations

Type	Target label	Target concept
related	One-Way Street	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000002

One-Way Street

Semantic relations

Type	Target label	Target concept
narrower	Dynamic One-Way Street	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000014
related	Closure Declaration	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000005
related	Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000006

Road Section

Semantic relations

Type	Target label	Target concept
related	Direction of Travel	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000004

Direction of Travel

Semantic relations

Type	Target label	Target concept
related	One-Way Street	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000002

Closure Declaration

Semantic relations

Type	Target label	Target concept
related	Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000006

Indication

Semantic relations

Type	Target label	Target concept
narrower	Start Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000008
narrower	Static Sign	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000007
narrower	End Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000009
related	Advance Warning Sign	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000011
related	Supplementary Sign	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000010

Static Sign

Semantic relations

Type	Target label	Target concept
broader	Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000006

Start Indication

Semantic relations

Type	Target label	Target concept
broader	Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000006

End Indication

Semantic relations

Type	Target label	Target concept
broader	Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000006

Supplementary Sign

Semantic relations

Type	Target label	Target concept
related	Exception	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000012

Exception

Semantic relations

Type	Target label	Target concept
related	Closure Declaration	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000005

Validity

Semantic relations

Type	Target label	Target concept
related	Supplementary Sign	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000010

Dynamic One-Way Street

Semantic relations

Type	Target label	Target concept
broader	One-Way Street	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000002
related	Validity	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000013

Traffic Order

Semantic relations

Type	Target label	Target concept
related	One-Way Street	https://ndwnu.github.io/its-gegevenscatalogus/id/758dc0d0-738b-4cff-9cf1-7f162258a064/concept-2603010000002